

Contents

1	AIX TradeNet	1
1.1		2
1.2	1.	2
	1.2.1 1.1	2
	1.2.2 1.2	2
	1.2.3 1.3	2
1.3	2.	3
	1.3.1 2.1	3
	1.3.2 2.2	3
	1.3.3 2.3	3
	1.3.4 2.4	4
1.4	3. API	4
	1.4.1 3.1	4
	1.4.2 3.2 API	4
	1.4.3 3.3 API	5
	1.4.4 3.4 API	5
	1.4.5 3.5	6
	1.4.6 3.6 Webhook	6
1.5	4. TradeNet	6
	1.5.1 4.1 AIX Chain	6
	1.5.2 4.2	7
	1.5.3 4.3	7
	1.5.4 4.4 DID	8
1.6	5.	8
	1.6.1 5.1	8
	1.6.2 5.2	9
	1.6.3 5.3	9
	1.6.4 5.4	10
	1.6.5 5.5	10
1.7	6.	11
	1.7.1 6.1	11
	1.7.2 6.2	11
	1.7.3 6.3	12
	1.7.4 6.4	12
1.8	7.	13
	1.8.1 7.1 API	13
	1.8.2 7.2	14
	1.8.3 7.3	15
1.9		16

1 AIX TradeNet

1.1

AIX TradeNet	WeChat Work	API	CRM	Web3
TradeNet	TradeNet	DID		
TradeNet	Web3	DID		

1.2 1.

1.2.1 1.1

Web3		AIX TradeNet	TradeNet
TradeNet	1.8	1200	TradeNet
	TradeNet	DID	
	TradeNet		TradeNet
DID			

1.2.2 1.2

	60%
TradeNet	” ” Web3
DID	GDPR
TradeNet	

1.2.3 1.3

AIX Chain	DID	AIX Chain	TradeNet	PoS	Solidity
AIX Chain	DID	W3C DID			
	TradeNet	API			
	API	API	TradeNet	Web	
TradeNet	SDK		TradeNet		
API					

1.3 2.

1.3.1 2.1

AIX TradeNet

API

DDoS

API

TradeNet

PostgreSQL

MongoDB

InfluxDB

Redis

AIX Chain

AIX Chain

1.3.2 2.2

PostgreSQL
API

DID

TradeNet

DID

DID

DID
AIX Chain

SDK

AIX Chain

1.3.3 2.3

Nginx
AES

IP

API

Kong
IP

RESTful API

WebSocket

OAuth 2.0
JWT
RBAC

TradeNet

OAuth
ABAC

TradeNet

API

ID

TradeNet DID

TradeNet

Redis RabbitMQ

BPMN 2.0

API

API

TradeNet

TradeNet

TradeNet

CQRS

CPU

Prometheus

Grafana

PromQL

Jaeger Zipkin

1.3.4 2.4

Node.js	TypeScript	Python	Node.js	I/O	API	WebSocket	I/O
Python			gRPC				
Web3.js	ethers.js	AIX Chain	Web3.js	API		ethers.js	
Rust							
PostgreSQL		MongoDB		Redis		Pub/Sub	Elasticsearch
RabbitMQ			Kafka		Redis Stream		
Docker		Kubernetes		Helm	Kubernetes		
DevOps	GitLab CI/CD	Terraform		Ansible	Prometheus + Grafana		
	ELK Stack	Elasticsearch + Logstash + Kibana					

1.4 3. API

1.4.1 3.1

API	API	RESTful	HTTPS
JSON			
API	API		
API			
CorpID	AgentId	Secret	access_token
cret	access_token	Access_token	API
		URL	AES

1.4.2 3.2 API

API

Access Token GET <https://qyapi.weixin.qq.com/cgi-bin/gettoken?corpid=ID&corpsecret=SECRET>
 API API access_token access_token

GET https://qyapi.weixin.qq.com/cgi-bin/user/getuserinfo?access_token=ACCESS_TOKEN&code=CODE
 OAuth code UserId

GET https://qyapi.weixin.qq.com/cgi-bin/user/get?access_token=ACCESS_TOKEN&userid=USERID

API

GET https://qyapi.weixin.qq.com/cgi-bin/department/list?access_token=ACCESS_TOKEN&id=ID

GET https://qyapi.weixin.qq.com/cgi-bin/user/simplelist?access_token=ACCESS_TOKEN&department=DEPARTMENT_ID

API

GET https://qyapi.weixin.qq.com/cgi-bin/externalcontact/list?access_token=ACCESS_TOKEN&userid=USERID

GET https://qyapi.weixin.qq.com/cgi-bin/externalcontact/get?access_token=ACCESS_TOKEN&external_userid=EXTERNAL_USERID

GET https://qyapi.weixin.qq.com/cgi-bin/externalcontact/get_follow_user_list?access_token=

API

POST https://qyapi.weixin.qq.com/cgi-bin/message/send?access_token=ACCESS_TOKEN

POST https://qyapi.weixin.qq.com/cgi-bin/message/update_taskcard?access_token=ACCESS_TOKEN

API

POST https://qyapi.weixin.qq.com/cgi-bin/media/upload?access_token=ACCESS_TOKEN&type=TYPE
media_id

POST https://qyapi.weixin.qq.com/cgi-bin/media/uploading?access_token=ACCESS_TOKEN
URL

1.4.3 3.3 API

TradeNet	API	TradeNet	DID						
	change_external_contact		add_external_contact					externalcontact/get	
API	external_userid								
	TradeNet	DID	DID		DID	DID	AIX Chain	DID	DID
		external_userid	DID						
								externalcontact/get_corp_tag_list	API
	externalcontact/mark_tag	API							
TradeNet		TradeNet	"VIP "		" "			DID	
	500		externalcontact/group_chat/list	API				externalcontact/group_chat/get	
API									
TradeNet	" "	" "	" "		DID				
			externalcontact/get_user_behavior_data					externalcontact/get_group_msg_resul	
	TradeNet								

1.4.4 3.4 API

	Markdown		@						
mpnews			HTML						
news			URL						
/ / /		20MB	JPG PNG		AMR		MP4		

TradeNet

1.4.5 3.5

Access Token	Access token	API	CorpID	Secret	Secret	Access
token	7200	2	access token			
	access token	access token		access token	API	
	AES	Token	EncodingAESKey	Token	EncodingAESKey	
1.	EncodingAESKey	AES	2.	Token	3.	
URL	4.	5.	EncodingAESKey	6.		

IP	IP	IP	IP
JSAPI	JSAPI	URL jsapi_ticket	JSAPI
JSAPI			

1.4.6 3.6 Webhook

Webhook

”	”	URL Token	EncodingAESKey	URL	HTTPS		
URL							
URL	GET	URL	msg_signature	timestamp	nonce	echostr	To-
ken	timestamp	nonce	echostr				
	POST	URL	XML	JSON			

- change_external_contact
- change_external_chat
- change_external_tag
- msgaudit_notify
- open_approval_change

200

3

1.5 4. TradeNet

1.5.1 4.1 AIX Chain

AIX Chain	TradeNet	EVM	AIX Chain	TradeNet		
	AIX Chain	RPC			Merkle	
RPC	AIX Chain	JSON-RPC	eth_getBalance	eth_sendTransaction	eth_call	eth_ge
Web3						

WebSocket	AIX Chain	WebSocket	WebSocket
AIX Chain			
	AIX Chain	JSON-RPC	

1.5.2 4.2

TradeNet	DID		
Call	Gas		
Transaction			
1. nonce	nonce	nonce	
2. Gas	eth_estimateGas	Gas	Gas Gas
3. Gas Price	Gas Price	EIP-1559	
4. ABI	4		
5.			
6. eth_sendRawTransaction			
7.			
ABI	ABI Application Binary Interface	ABI	ABI
ABI			

1.5.3 4.3

Event			
Solidity	event	emit	indexed
event CustomerRegistered(bytes32 indexed did, string externalUserId, uint256 timestamp);			
Web3	DID	ID	
			Saga

1.5.4 4.4 DID

DID Decentralized Identifier TradeNet TradeNet DID

DID DID JSON-LD DID DID

```
{
  "@context": "https://www.w3.org/ns/did/v1",
  "id": "did:aix:0x1234567890abcdef",
  "verificationMethod": [{
    "id": "did:aix:0x1234567890abcdef#keys-1",
    "type": "EcdsaSecp256k1VerificationKey2019",
    "controller": "did:aix:0x1234567890abcdef",
    "publicKeyHex": "02b97c30de767f084ce3080168ee293053ba33b235d7116a3263d29f1450936b71"
  }],
  "authentication": ["did:aix:0x1234567890abcdef#keys-1"],
  "service": [{
    "id": "did:aix:0x1234567890abcdef#wechat",
    "type": "WeChatService",
    "serviceEndpoint": "https://work.weixin.qq.com/external_user/wm123456"
  }]
}
```

DID

1. secp256k1
2. DID DID did:aix: +
3. DID
4. DID DID AIX Chain Gas
5. DID

DID DID external_userid did corpid DID DID

DID DID DID

VC DID "VIP "

DID DID DID DID DID DID DID

1.6 5.

1.6.1 5.1

TradeNet

Gateway Service API Node.js

Express Redis

Protocol Service TradeNet XML TradeNet JSON-RPC

Identity Service	TradeNet DID	TradeNet	DID
OAuth TradeNet			
Transform Service	TradeNet	TradeNet	TradeNet
Cache Service	access_token	DID	Redis
Config Service	CorpID Secret AIX Chain	RPC	

1.6.2 5.2

TradeNet

cron

TradeNet TradeNet

TradeNet - external_userid TradeNet externalUserId - name
TradeNet displayName - follow_userid TradeNet followerId

- - TradeNet -

Saga

1.6.3 5.3

” ” ” ” NLP

VIP

” ”

RabbitMQ Kafka Redis Stream

ID

1.6.4 5.4

TradeNet

CPU

1.6.5 5.5

5

1 2 4 8

thundering herd

HTTP

ELK Stack

Grafana

1.7 6.

1.7.1 6.1

4

- Kubernetes DevOps - - API - AIX Chain
- - - API -

6

DID

- - DID - - API
- - - DID - SDK -

6

- - - - -
- - - - -

4

- - - - -
- - - - -

1.7.2 6.2

M1 2

API AIX Chain

- - AIX Chain - API < 200ms

M2 10

DID

- DID - 90% - 80%

M3 16

- - - 1000

M4 20

- - - 10000 -

1.7.3 6.3

- 1
- 1
- 4
- 2
- 2
- DevOps 1 CI/CD
- 2
- 1
- UI 1

- Kubernetes 3 16C64G - PostgreSQL 8C32G - Redis 4C16G - RabbitMQ 4C16G

- AIX Chain 1 4C16G

- Kubernetes 10 16C64G - PostgreSQL 16C64G - Redis 8C32G - RabbitMQ 8C32G - AIX Chain 2 8C32G - CDN

-
- AWS/ /
- Datadog/
- ELK Stack

1.7.4 6.4

API API

AIX Chain Gas Gas Price Layer 2

1.8 7.

1.8.1 7.1 API

POST /api/v1/customers/sync

```
{
  "externalUserId": "wm123456789",
  "corpId": "ww123456789",
  "userId": "zhangsan",
  "name": " ",
  "avatar": "https://...",
  "tags": ["VIP", " "]
}
```

```
{
  "code": 0,
  "message": "success",
  "data": {
    "customerId": "CUST123456",
    "did": "did:aix:0xabcdef...",
    "syncStatus": "completed"
  }
}
```

POST /api/v1/messages/send

```
{
  "externalUserIds": ["wm123456789"],
  "msgType": "text",
}
```

```

    "content": {
      "text": "      "
    }
  }

{
  "code": 0,
  "message": "success",
  "data": {
    "messageId": "MSG123456",
    "sendStatus": "sent"
  }
}

```

POST /api/v1/contracts/call

```

{
  "contractAddress": "0x123456...",
  "method": "registerCustomer",
  "params": {
    "did": "did:aix:0xabcdef...",
    "metadata": "{\"name\":\" \" \"}"
  }
}

{
  "code": 0,
  "message": "success",
  "data": {
    "transactionHash": "0xabcdef...",
    "status": "pending"
  }
}

```

1.8.2 7.2

```

wechat_work:
  corp_id: "ww1234567890abcdef"
  agent_id: 1000002
  secret: "your-agent-secret-here"
  contact_secret: "your-contact-secret-here"
  token: "your-token-here"
  encoding_aes_key: "your-encoding-aes-key-here"
  callback_url: "https://your-domain.com/api/wechat/callback"

```

AIX Chain

```
aix_chain:
  network: "mainnet" # "testnet"
  rpc_url: "https://rpc.aixchain.io"
  websocket_url: "wss://ws.aixchain.io"
  chain_id: 1314
  gas_price: "200000000000" # 20 Gwei
  contracts:
    did_registry: "0x1234567890abcdef..."
    customer_manager: "0xabcdef1234567890..."
    trade_contract: "0xfedcba0987654321..."
```

```
database:
  postgresql:
    host: "localhost"
    port: 5432
    database: "tradenet"
    username: "tradenet_user"
    password: "your-password"
    pool_size: 20
```

```
redis:
  host: "localhost"
  port: 6379
  password: "your-password"
  db: 0
```

1.8.3 7.3

0	Success	
10001	Invalid Parameter	
10002	Unauthorized	
10003	Forbidden	
10004	Not Found	
10005	Internal Error	
20001	WeChat API Error	API
20002	Access Token Expired	Access Token
20003	Callback Verify Failed	
30001	Blockchain Error	
30002	Transaction Failed	
30003	DID Not Found	DID
30004	Insufficient Gas	Gas
40001	Customer Not Found	
40002	Customer Already Exists	
40003	Sync Failed	

50001	Message Send Failed
50002	Rate Limit Exceeded

1.9

AIX TradeNet	TradeNet
AIX Chain	
Web3	TradeNet

v1.0
2024
AIX TradeNet